🏆 RS003 Week 16 — English Worksheet

Topic: Polish — Missiles, Sprites, High Score, and Show & Tell · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This is the final week. In this worksheet you think about and explain the **polish** that makes a game feel finished: a second weapon (**missiles** with the M key) that hits harder, a **high score** that remembers your best, a restart key, and real **PNG sprites** loaded from image files. Then you present and explain the finished game you built across the whole course.

> 🧠 Words to know: **missile**, **sprite**, **image.load**, **high score**, **restart**

1 · Predict 🔮

Read each piece of code. Write what you think it does. You do not need to run anything — just read and think.

```
ship_img = pygame.image.load("ship.png").convert_alpha()
enemy_img = pygame.image.load("enemy.png").convert_alpha()
screen.blit(ship_img, (ship_x - ship_img.get_width() // 2, ship_y))
```

These lines load picture files and put them on screen. What replaces your drawn rectangles and triangles?


```
if event.key == pygame.K_m:
    missiles.append([ship_x, ship_y - 20])
```

Pressing M fires a missile instead of a bullet. Why might a game want a second, stronger weapon?

```
high_score = 0
if event.key == pygame.K_r and game_over:
    if score > high_score:
        high_score = score
    score = 0
    game_over = False
```

Pressing R restarts the game. When does the high score update?

```
for mi in missiles[:]:
    if abs(mi[0] - e["x"]) < 30 and abs(mi[1] - e["y"]) < 30:
        score += e["points"] * 2
```

A missile gives `e["points"] * 2`. How many points does a 3-point meteor give when a missile hits it?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Write the fixed code, then explain why the original was wrong.

Bug A — The sprite is supposed to show your `ship.png` picture, but the program crashes saying it cannot find the file.

```
ship_img = pygame.image.load("ship.png")
```

Hint: the picture file must be saved in the **same folder** as your code, with the exact name. Also use `.convert_alpha()` so the see-through background stays see-through.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The high score should remember your best across rounds, but it resets to 0 every restart.

```
if event.key == pygame.K_r and game_over:
    high_score = 0
    score = 0
    game_over = False
```

Hint: before resetting, compare the score to the best so far.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — After the timer hits 0, the player should not be able to keep flying or firing, but the game keeps playing.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]:
    ship_x -= SPEED
```

Hint: movement should only run while the game is not over.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code 📖

Read the central loop from this week's finished game. The full program is long, so this is the key part — the part that fires both weapons, loads the sprites, and handles restart. Read it carefully and answer the questions below. You do not need to run it.

```
ship_img = pygame.image.load("ship.png").convert_alpha()
enemy_img = pygame.image.load("enemy.png").convert_alpha()

for event in pygame.event.get():
    if event.type == pygame.KEYDOWN:
        if event.key == pygame.K_SPACE and not game_over:
            bullets.append([ship_x, ship_y - 20])
        if event.key == pygame.K_m and not game_over:
            missiles.append([ship_x, ship_y - 20])
        if event.key == pygame.K_r and game_over:
            if score > high_score:
                high_score = score
            score = 0
            enemies.clear()
            start_time = pygame.time.get_ticks()
            game_over = False

for b in bullets: b[1] -= 8
for mi in missiles: mi[1] -= 4

for e in enemies[:]:
    for b in bullets[:]:
        if abs(b[0] - e["x"]) < 22 and abs(b[1] - e["y"]) < 22:
            score += e["points"]
            if e in enemies: enemies.remove(e)
            if b in bullets: bullets.remove(b)
    for mi in missiles[:]:
        if abs(mi[0] - e["x"]) < 30 and abs(mi[1] - e["y"]) < 30:
            score += e["points"] * 2
            if e in enemies: enemies.remove(e)
            if mi in missiles: missiles.remove(mi)

screen.blit(enemy_img, (e["x"] - enemy_img.get_width() // 2, e["y"] -
enemy_img.get_height() // 2))
screen.blit(ship_img, (ship_x - ship_img.get_width() // 2, ship_y))
```

Which line loads the ship picture, and what does `.convert_alpha()` do for it?

The bullet moves with `b[1] -= 8` and the missile with `mi[1] -= 4`. Which one moves faster up the screen, and how do you know?

A bullet hit gives `score += e["points"]` but a missile hit gives `score += e["points"] * 2`. Why is the missile worth more?

The R key only works `if ... game_over`. Why should restart only work when the game is over?

`screen.blit(ship_img, ...)` puts a picture on screen. How is `blit` different from `pygame.draw.rect`?

4 · Explain Your Lesson Code 🎥

This is your **show & tell**. Present the finished game **you** built across this whole course. Make a phone video: show the game running from start to game over to restart, and explain the key parts of your code out loud. Try to use these words: **missile**, **sprite**, **image.load**, **high score**, **restart**.

> 1. Play a full round, firing both bullets and missiles, and let the timer run out.
> 2. Show the game-over screen and your high score, then press R and show it restart.
> 3. Point to the line that uses `image.load` and explain how you loaded your PNG sprites.
> 4. Tell us the part of your code you are most proud of and one bug you fixed.

Write what you will say in your video. Plan it here before you record.

Submit ✅

Send this worksheet + your finished-game presentation video to teacher on KakaoTalk.